



Product Analytics

Product Analytics (E-Wallet)

Business Problem

Bill payment success rate has dropped significantly since August 2025.

Business Objective

Identify root causes of the declining success rate.
Provide actionable recommendations to restore and improve performance.





● Dataset Overview (July 1 – Aug 31, 2023)

- **transactions.csv** – transaction history (success/failure).
- **sessions.csv** – session metadata (duration, connection, app version).
- **event_logs.csv** – detailed user events during payment flow.
- **dim_user.csv** – user demographics, device, loyalty tier.
- **dim_event.csv** – event definitions (navigation, authentication, payment).
- **dim_error_categories.csv** – multi-level error classification.

● Context

- Several app updates occurred during the analysis window (v3.9.6 – v3.9.9) introducing **new authentication, redesigned payment flow, and gateway**.
- Need to assess whether **product changes, user/device segments, or external integrations** caused the decline.

Framework: B.A.D Approach



Business Context Mastery

- **KPI:** Success Rate = Successful Transactions ÷ Total Transactions.
- **Stakeholders:** Product, Engineering, BI/Data, Support, Management.



Analytical Strategy

- **User-related factors:** device type, activity level, loyalty tier.
- **System factors:** app version, OS version, connection quality.
- **Process factors:** authentication (OTP/3DS), merchant integrations.
- **External factors:** payment methods, promotions, time-of-day effects.

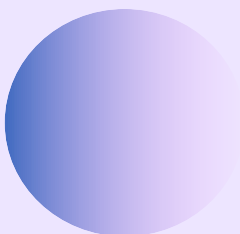


Data-Driven Execution

- Trend & funnel analysis.
- Error root cause mapping.
- User/device segmentation.
- Cohort comparison (July vs August).

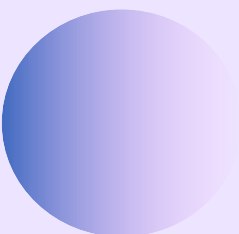
Step 1. Define Key Metrics

- Success Rate % by day (overall vs billing).
- Drop-off rate per funnel step.



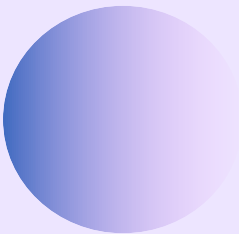
Step 3. Error Analysis

- Map failed transactions to **dim_error_categories**.
- Prioritize by weight/frequency.



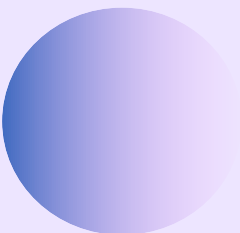
Step 5. Synthesis & Insights

- Link technical updates with success rate decline.
- Consolidate into actionable recommendations.



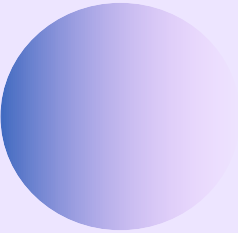
Step 2. Funnel Analysis

- Compare July vs August in payment path.
- Identify where failures spike (e.g., authentication, confirmation).



Step 4. Segmentation

- Device/platform differences (Android, iOS, Web).
- User tier & activity level.
- Connection quality & app version.



Bill-payment success rate plunged in mid-August (–25–30 pp vs July) and only partially recovered by early September

From a July high (~92%), the 7-day SR collapsed to ~60–65% mid-August and recovered only to ~70–75% by early September; the **billing series tracks it one-for-one**, ruling out data-scope bias.

Bill Payment Success Rate Trend

● Overall Success Rate (7D Rolling Avg) ● Billing Success Rate

Peak
~92%
(Jul)

Trough
~60–65%
(mid-Aug)

Δ from July: –25–30 pp

Thg7 2023

Thg8 2023

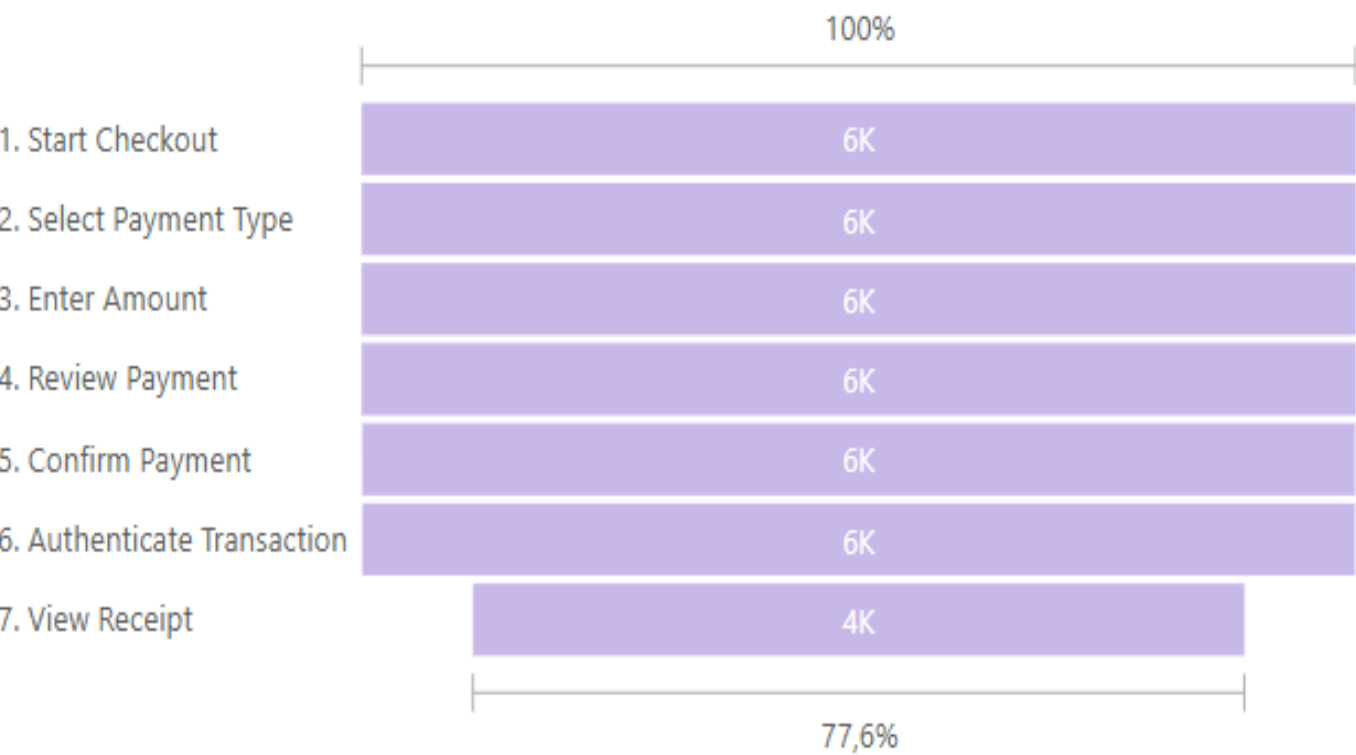
Thg9 2023

No single step collapsed; final completion is ~77.6% of “Start Checkout”. Step-by-step conversion is stable across versions (3.9.5–3.9.9)

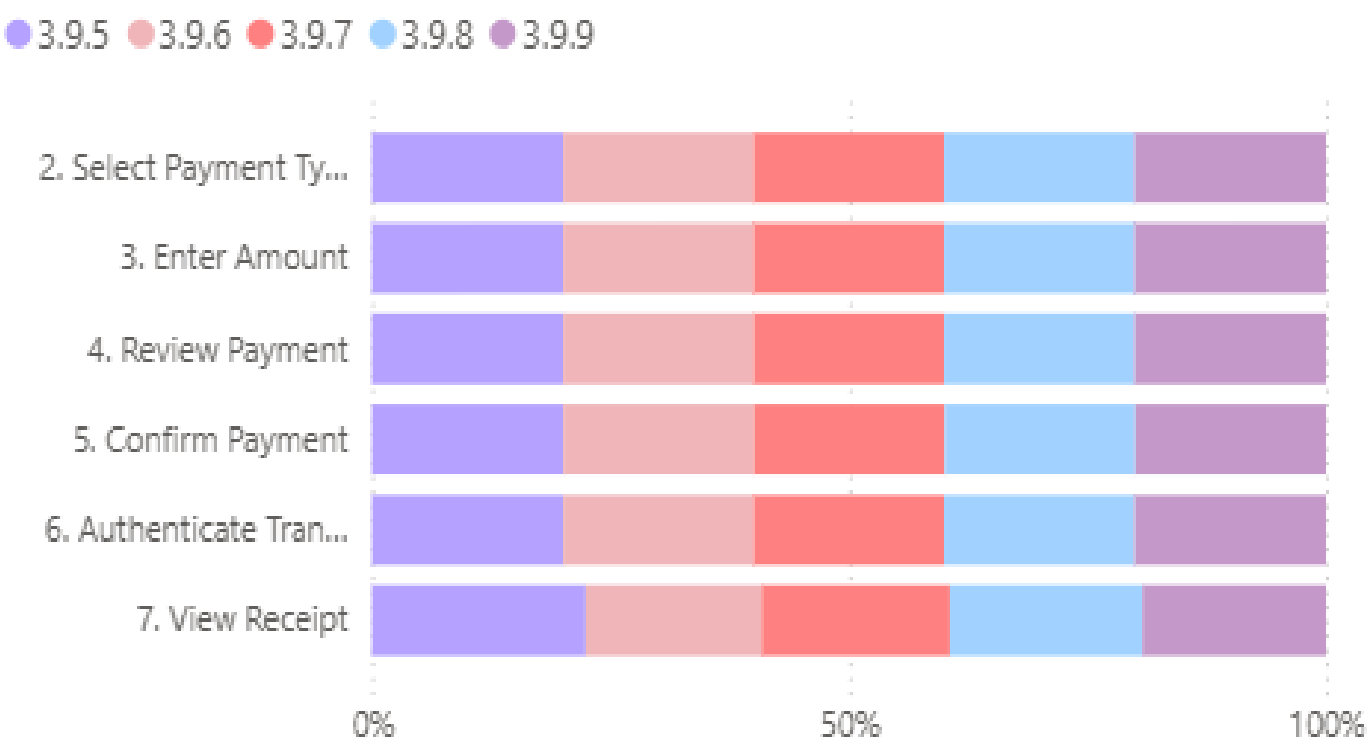
Not a UX step drop-off → investigate errors & version/platform.

Each step’s conversion from the previous step remains stable, with no version-specific dip for 3.9.6–3.9.7 at any individual stage.

Bill Payment Funnel — Stage Completion (Billing)



Percent of Previous by Stage & app_version

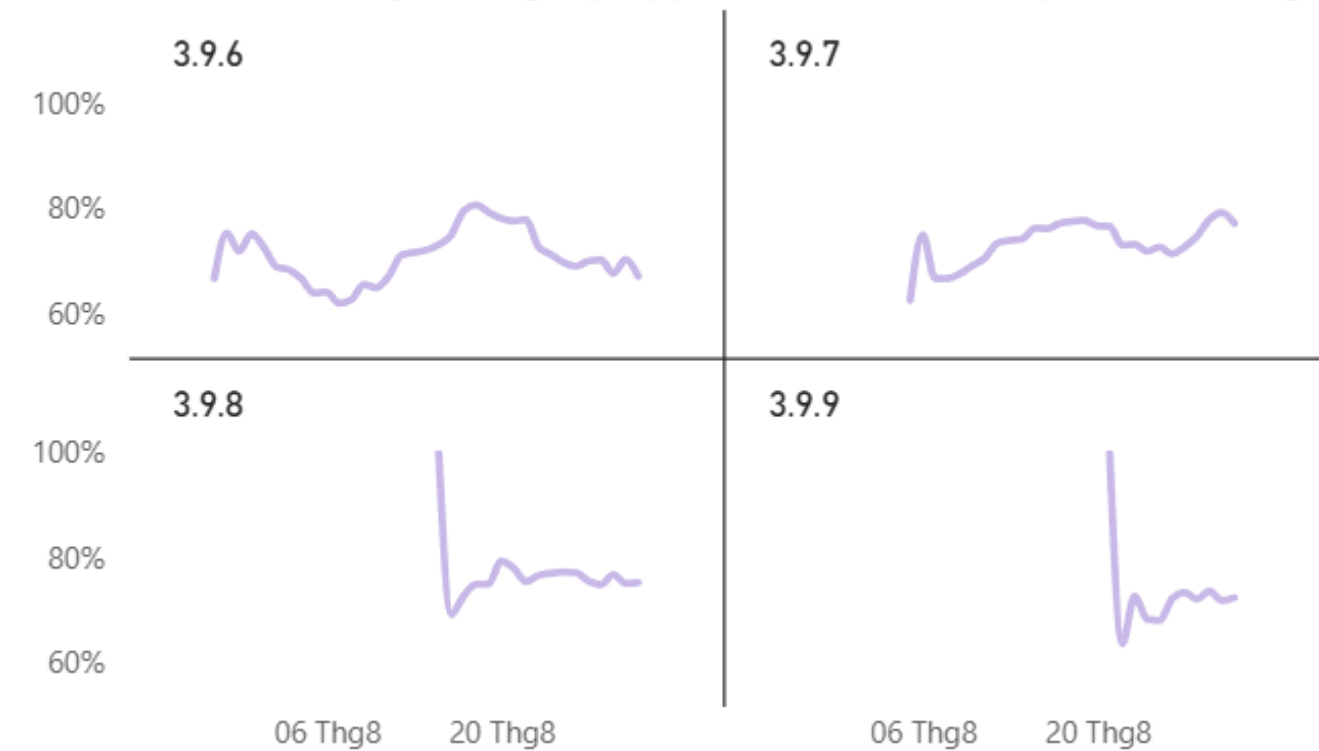


The August decline is version-driven—releases 3.9.6–3.9.7 underperform (SR ~60–65% mid-Aug, ending ~70%) while 3.9.8–3.9.9 hold ~70–80%

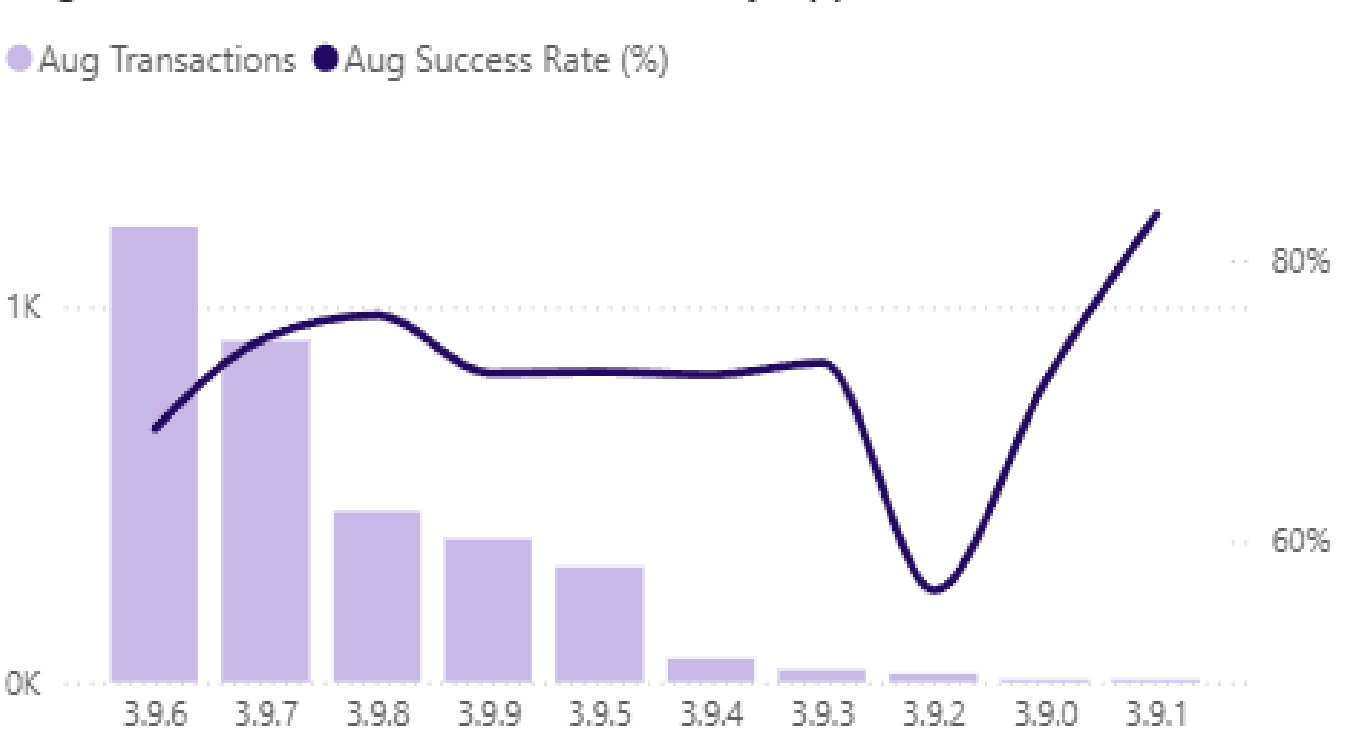
The lines for v3.9.6–3.9.7 plunge from ~90% in July to ~60–65% mid-August and recover only to ~70%, whereas v3.9.8–3.9.9 dip less and finish ~75–80%, showing the degradation is concentrated in specific releases.

v3.9.6–3.9.7 carry the largest August transaction share and also post the lowest SR (~0.60–0.70), confirming these high-volume, low-performance cohorts are the primary drivers of the overall drop.

Success Rate (7-day rolling) by App Version — Jul–Sep 2025 (Billing)



August Transactions & Success Rate by App Version

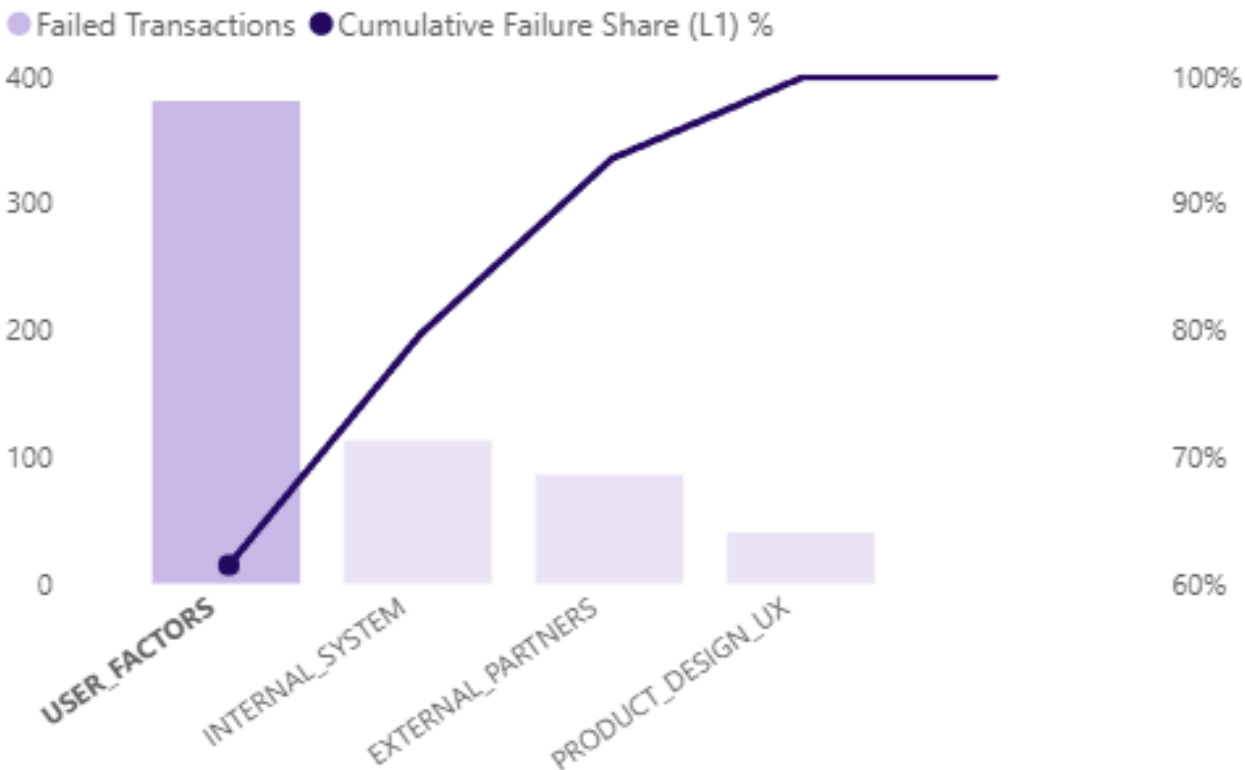


Bill-payment failures are overwhelmingly user/authentication-driven; by the third L1 category the Pareto already explains ~90% of all fails.

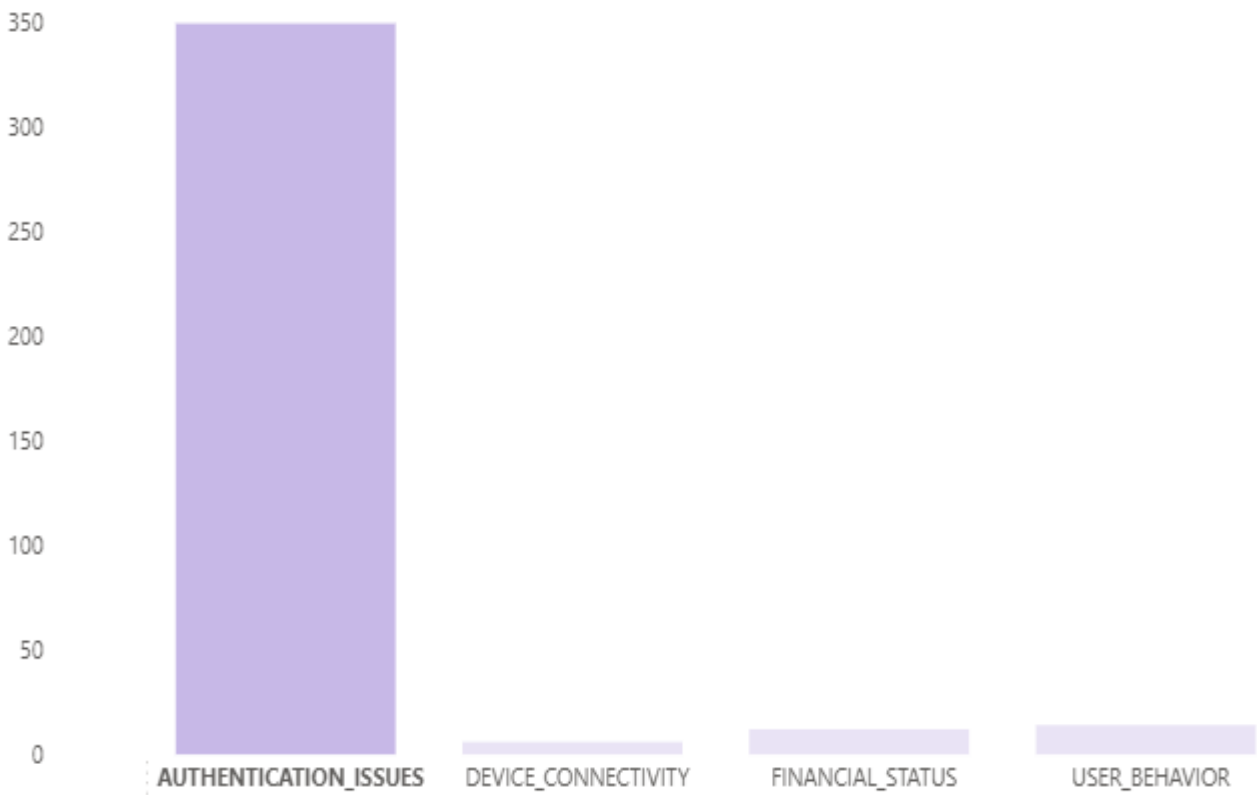
USER_FACTORS lead (~60–65% of fails), followed by INTERNAL_SYSTEM (~20–25%) and EXTERNAL_PARTNERS (~10–12%); the cumulative line hits ~90% by the third bar.
→ Most failures originate on the user/auth path rather than a single partner or UX issue.

AUTHENTICATION_ISSUES account for >80–90%, while DEVICE_CONNECTIVITY / FINANCIAL_STATUS / USER_BEHAVIOR are minor tails → the problem narrows to the auth flow.

Failure Pareto — Level 1 (Aug 2025, v3.9.6–3.9.7)



USER_FACTORS Breakdown — Level 2 (Aug 2025, v3.9.6–3.9.7)

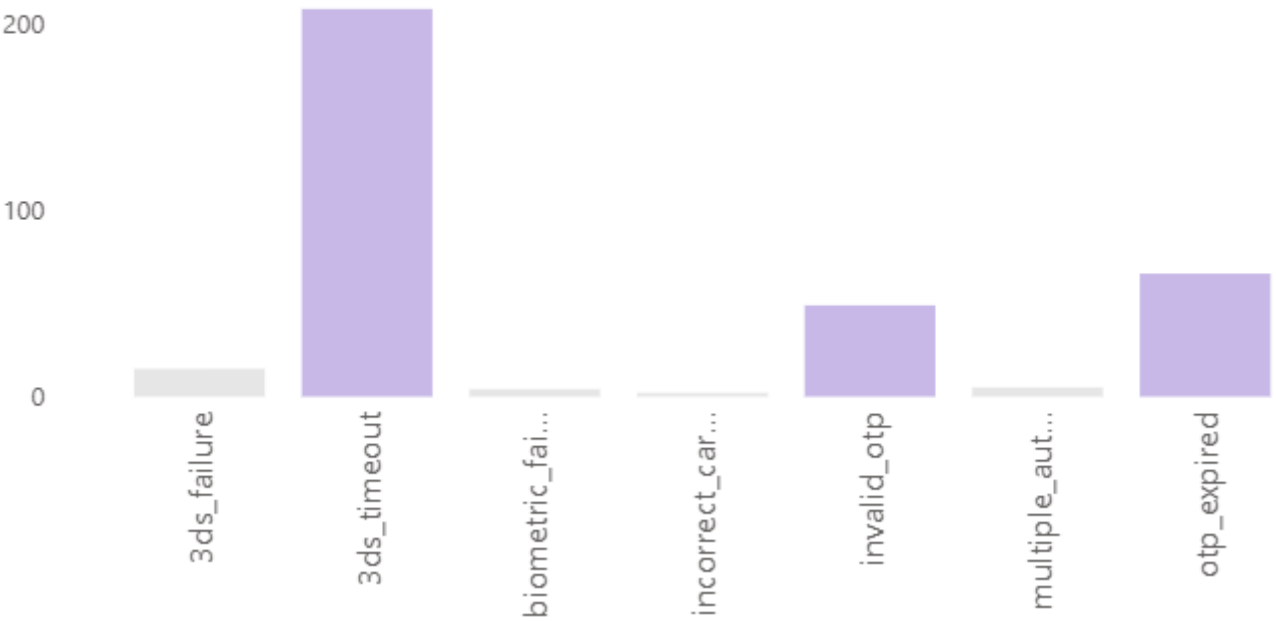


SR drop is authentication-driven—3DS timeouts are the main cause, worse on weak connections; OTP expired/invalid is secondary.

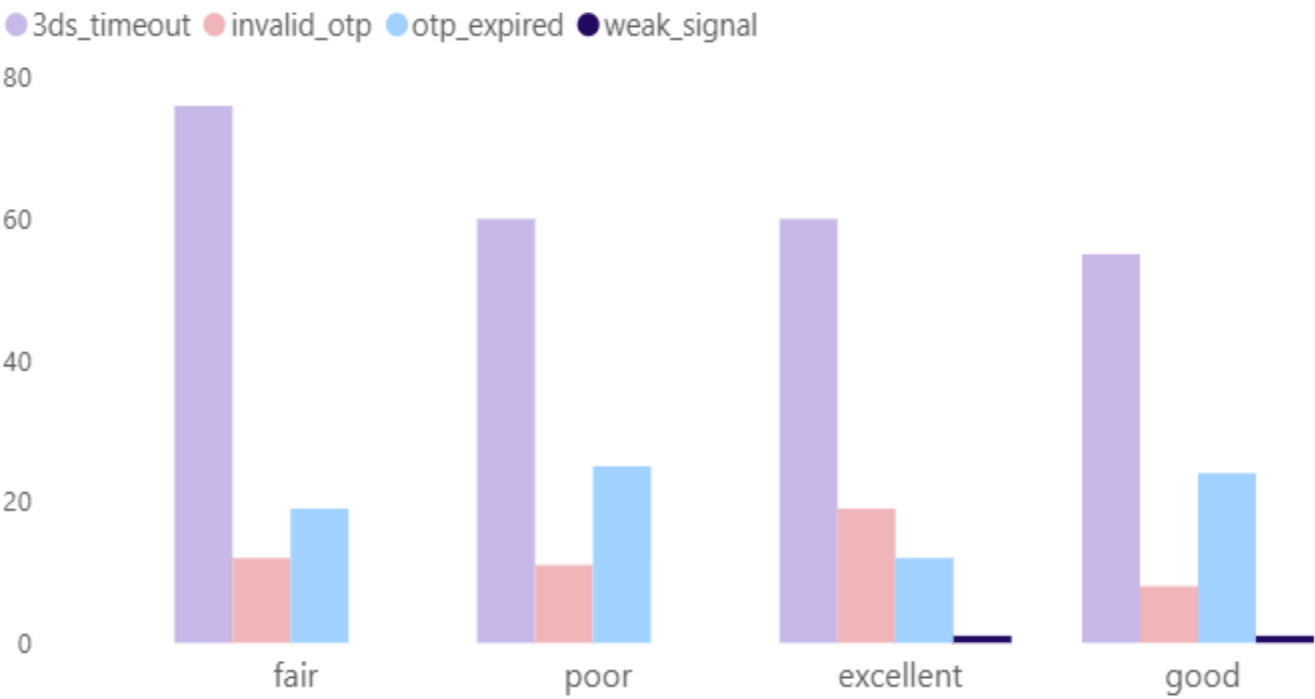
Timeouts dominate authentication failures, with OTP issues (expired/invalid) next; together they explain the bulk of August losses on v3.9.6–3.9.7.

3DS timeouts spike from good → fair/poor connections → network-sensitive 3DS is the bottleneck.

Authentication Reasons — Level 3 (Aug 2025, v3.9.6–3.9.7)



Failed auth reasons * connection quality



What explains the drop?

- When: mid-Aug
- Where: v3.9.6–3.9.7
- What: auth **timeouts** + OTP issues; worse on **weak connections**.
- What not: no single funnel-step collapse; decline broad across products
- Why (likely): 3DS/OTP auth flow & gateway latency regressions in those builds.

Quick Wins (7–14 days)

● 3DS/Gateway

- Increase timeout slightly + 1 retry
- Traffic-shaping at peak hours
- Fallback to OTP when 3DS p95 latency > threshold

● OTP UX

- Extend expiry +30–60s, resend after 15s
- Show countdown; persist input on app resume

● Connectivity

- Pre-auth network check & prompt “Switch to stronger signal”

Monitoring & Guardrails

● Monitors

- SR 7D (overall, billing, by version/method/reason)
- Auth timeout rate, OTP invalid/expired by version
- 3DS latency p95/p99

● SLO & Alerts

- Gateway SLO: p95/p99 latency
- Alerts: 3DS latency ↑, timeout share ↑, SR $\Delta < -3pp$ d/d
- Auto page/Slack to the on-call

● Release Guardrails

- Block rollout if SR < baseline -5pp
- Canary per build, auto-rollback
- Observability: log error_reason + app_version + partner

Thank You!

Get In Touch With Me



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